Homework 8

Shanshan Cao

7.3.1

Set $\psi = \sum_{n=0}^{\infty} a_n y^n$, we have $\psi'' = \sum_{n=0}^{\infty} a_n n(n-1) y^{n-2} = \sum_{n=2}^{\infty} a_n n(n-1) y^{n-2}$.

Therefore,

$$
\begin{aligned}
0 &= \psi'' + (2\epsilon - y^2)\psi \\
&= \sum_{n=2}^{\infty} a_n n(n-1) y^{n-2} + 2\epsilon \sum_{n=0}^{\infty} a_n y^n - y^2 \sum_{n=0}^{\infty} a_n y^n \\
&= \sum_{n=0}^{\infty} a_{n+2}(n+2)(n+1) y^n + 2\epsilon \sum_{n=0}^{\infty} a_n y^n - \sum_{n=2}^{\infty} a_{n-2} y^n \\
&= (2a_2 + 2\epsilon a_0) + (6a_3 + 2\epsilon a_1) y + \sum_{n=2}^{\infty} \left[a_{n+2}(n+2)(n+1) + 2\epsilon a_n - a_{n-2} \right] y^n
\end{aligned}
$$

$$
\Rightarrow \begin{cases} a_2 = -\epsilon a_0 \\ a_3 = -\frac{1}{3}\epsilon a_1 \\ a_{n+2} = \dfrac{a_{n-2} - 2\epsilon a_n}{(n+2)(n+1)}, \quad n \geq 2 \end{cases}
$$

where a_0 and a_1 are arbitrary constants (may be determined by normalization).

7.3.2

Eqn. (7.3.15) $\quad C_{n+2} = C_n \dfrac{2n+1-2\epsilon}{(n+2)(n+1)}$

For H_3, $\epsilon_3 = 3 + \frac{1}{2} = \frac{7}{2}$

$$C_3 = C_1 \frac{2\times 1 + 1 - 7}{(1+2)(1+1)} = -\frac{2}{3} C_1$$

$$C_5 = C_3 \frac{2\times 3 + 1 - 7}{(3+2)(3+1)} = 0$$

$$C_7 = C_9 = C_{11} = \cdots = 0$$

If we choose $C_1 = -12$, $C_2 = 0$, we have

$C_3 = 8$, $C_4 = C_6 = C_8 = \cdots = 0$.

Therefore, $H_3(y) = -12y + 8y^3$.

For H_4, $\epsilon_4 = 4 + \frac{1}{2} = \frac{9}{2}$

$$C_2 = C_0 \frac{0+1-9}{2\times 1} = -4C_0$$

$$C_4 = C_2 \frac{4+1-9}{4\times 3} = -\frac{1}{3}C_2 = \frac{4}{3}C_0$$

$$C_6 = C_4 \frac{8+1-9}{6\times 5} = 0$$

$$C_8 = C_{10} = C_{12} = \cdots = 0$$

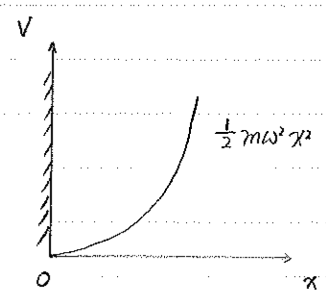
If we choose $C_0 = 12$, $C_1 = 0$, we have

$$C_2 = -3, \quad C_4 = 16, \quad C_3 = C_5 = C_7 = \cdots = 0.$$

Therefore, $H_4(y) = 12 - 48y^2 + 16y^4$.

7.3.6

This is similary to the regular harmonic oscillator problem, but with only the odd solutions, so that



$\psi(0) = 0$, (boundary condition)

required by $V(x \leq 0) = \infty$.

The normalization factor changes:

For the regular harmonic oscillator, $\int_{-\infty}^{\infty} \psi^* \psi \, dx = 1$;

but here, $\int_0^{\infty} \psi'^* \psi' dx = 1$.

Therefore $\psi'_n = \sqrt{2}\,\psi_n$, where ψ_n is the eigenfunctions of the regular harmonic oscillator, and is given by Eqn. (7.3.22).

To sum up, for this problem,

$$\psi_n = \sqrt{2}\left\{\frac{m\omega}{\pi\hbar\, 2^{2(2n+1)}[(2n+1)!]^2}\right\}^{\frac{1}{4}} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) H_{2n+1}\left[\left(\frac{m\omega}{\hbar}\right)^{\frac{1}{2}} x\right],$$

$$E_n = \left(2n+1+\frac{1}{2}\right)\hbar\omega = \left(2n+\frac{3}{2}\right)\hbar\omega,$$

$$n = 0, 1, 2, 3, \cdots.$$

7.4.3

For a circular orbit:

$$F = \frac{\partial V}{\partial r} = k a r^{k-1}$$

$$F = m v^2 / r$$

$$\Rightarrow m v^2 / r = k a r^{k-1}$$

$$\Rightarrow m v^2 = k a r^k$$

$$\Rightarrow T = \frac{1}{2} m v^2 = \frac{1}{2} k a r^2 = \frac{k}{2} V.$$

Therefore, $c(k) = \frac{1}{2} k$.

For the quantum oscillator:

$$\langle T \rangle = \frac{1}{2m} \langle P^2 \rangle$$

$$\langle P^2 \rangle = -\frac{m\omega\hbar}{2} \langle n | (a^+ - a)^2 | n \rangle$$

$$= -\frac{m\omega\hbar}{2} \langle n | a^+a^+ - a^+a - aa^+ + aa | n \rangle$$

$$\langle n | a^+a^+ | n \rangle = \sqrt{n} \cdot \sqrt{n+1} \; \langle n-1 | n+1 \rangle = 0$$

$$\langle n | a^+a | n \rangle = \sqrt{n} \cdot \sqrt{n} \; \langle n-1 | n-1 \rangle = n$$

$$\langle n | aa^+ | n \rangle = \sqrt{n+1} \cdot \sqrt{n+1} \; \langle n+1 | n+1 \rangle = n+1$$

$$\langle n | aa | n \rangle = \sqrt{n+1} \cdot \sqrt{n} \; \langle n+1 | n-1 \rangle = 0$$

$$\Rightarrow \langle P^2 \rangle = \frac{m\omega\hbar}{2} (n + n + 1) = \frac{m\omega\hbar^2}{2} (2n+1)$$

$$\Rightarrow \langle T \rangle = \frac{\hbar\omega}{4} (2n+1)$$

$$\langle V \rangle = \frac{1}{2} m\omega^2 \langle X^2 \rangle$$

$$\langle X^2 \rangle = \frac{\hbar}{2m\omega} \langle n | (a + a^+)^2 | n \rangle$$

$$= \frac{\hbar}{2m\omega} \langle n | aa + aa^+ + a^+a + a^+a^+ | n \rangle$$

$$= \frac{\hbar}{2m\omega} (n + n + 1)$$

$$= \frac{\hbar}{2m\omega} (2n+1)$$

$$\Rightarrow \langle T \rangle = \frac{\hbar\omega}{4} (2n+1)$$

Therefore, $\langle T \rangle = \langle V \rangle$.

7.4.5.

(1) $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}\,|\psi(0)\rangle$

$$\hat{H}\,|0\rangle = \tfrac{1}{2}\hbar\omega\,|0\rangle$$

$$\hat{H}\,|1\rangle = \tfrac{3}{2}\hbar\omega\,|0\rangle$$

$$\Rightarrow |\psi(t)\rangle = \tfrac{1}{\sqrt{2}}\left(e^{-i\hat{H}t/\hbar}\,|0\rangle + e^{-i\hat{H}t/\hbar}\,|1\rangle\right)$$

$$= \tfrac{1}{\sqrt{2}}\left(e^{-\frac{i\omega t}{2}}\,|0\rangle + e^{-\frac{3i\omega t}{2}}\,|1\rangle\right).$$

(2) $\langle X(0)\rangle = \frac{1}{2}\left(\langle 0| + \langle 1|\right)\sqrt{\frac{\hbar}{2m\omega}}\,(a+a^{+})\,\big|\left(|0\rangle + |1\rangle\right)$

$$= \frac{1}{2}\cdot\sqrt{\frac{\hbar}{2m\omega}}\,\left(\left(\langle 0| + \langle 1|\right)\big|\,a+a^{+}\,\big|\left(|0\rangle + |1\rangle\right)\right)$$

$$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(\langle 0|a|1\rangle + \langle 1|a^{+}|0\rangle\right)$$

$$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\,(1+1)$$

$$= \sqrt{\frac{\hbar}{2m\omega}}.$$

$$\langle P(0)\rangle = \frac{1}{2}\cdot i\cdot\sqrt{\frac{m\omega\hbar}{2}}\,\left(\left(\langle 0| + \langle 1|\right)\big|\,a^{+}-a\,\big|\left(|0\rangle + |1\rangle\right)\right)$$

$$= \frac{i}{2}\sqrt{\frac{m\omega\hbar}{2}}\left(-\langle 0|a|1\rangle + \langle 1|a^{+}|0\rangle\right)$$

$$= \frac{i}{2}\sqrt{\frac{m\omega\hbar}{2}}\,(-1+1)$$

$$= 0.$$

$$\langle X(t)\rangle = \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(\langle 0|e^{\frac{i\omega t}{2}} + \langle 1|e^{\frac{3i\omega t}{2}}\right)\big|\,a+a^{+}\,\big|\left(e^{-\frac{i\omega t}{2}}|0\rangle + e^{-\frac{3i\omega t}{2}}|1\rangle\right)$$

$$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(e^{-i\omega t}\langle 0|a|1\rangle + e^{i\omega t}\langle 1|a^{+}|0\rangle\right)$$

$$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\left(e^{-i\omega t} + e^{i\omega t}\right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\cos\omega t.$$

$$\langle P(t)\rangle = \frac{i}{2}\sqrt{\frac{m\omega\hbar}{2}}\left(\langle 0|e^{\frac{i\omega t}{2}} + \langle 1|e^{\frac{3i\omega t}{\hbar}}\right)\big|\,a^{+}-a\,\big|\left(e^{-\frac{i\omega t}{2}}|0\rangle + e^{-\frac{3i\omega t}{2}}|1\rangle\right)$$

$$= \frac{i}{2}\sqrt{\frac{m\omega\hbar}{2}}\left(-e^{-i\omega t}\langle 0|a|1\rangle + e^{i\omega t}\langle 1|a^{+}|0\rangle\right)$$

$$= \frac{i}{2}\sqrt{\frac{m\omega\hbar}{2}}\left(e^{i\omega t} - e^{-i\omega t}\right)$$

$$= -\sqrt{\frac{m\omega\hbar}{2}}\sin\omega t.$$

(3) Ehrenfest's theorem :

$$\langle \dot{O} \rangle = \frac{1}{i\hbar} \langle [O, H] \rangle .$$

$$\langle \dot{X} \rangle = \frac{1}{i\hbar} \langle [X, H] \rangle$$

$$= \frac{1}{i\hbar} \langle [X, \frac{P^2}{2m} + \frac{1}{2} m\omega^2 X^2] \rangle$$

$$= \frac{1}{i\hbar} \cdot \frac{1}{2m} \langle [X, P^2] \rangle$$

$$= \frac{1}{i\hbar} \cdot \frac{1}{2m} \langle 2i\hbar P \rangle$$

$$= \frac{1}{m} \langle P \rangle \qquad (1)$$

$$\langle \dot{P} \rangle = \frac{1}{i\hbar} \langle [P, H] \rangle$$

$$= \frac{1}{i\hbar} \langle [P, \frac{P^2}{2m} + \frac{1}{2} m\omega^2 X^2] \rangle$$

$$= \frac{1}{i\hbar} \langle [P, \frac{1}{2} m\omega^2 X^2] \rangle$$

$$= \frac{1}{i\hbar} \langle -\frac{1}{2} m\omega^2 \cdot 2i\hbar \cdot X \rangle$$

$$= -m\omega^2 \langle X \rangle \qquad (2)$$

Combining (1) and (2), we have

$$\langle \ddot{X} \rangle = \frac{1}{m} \langle \dot{P} \rangle = -\omega^2 \langle X \rangle$$

$$\Rightarrow \langle \ddot{X} \rangle + \omega^2 \langle X \rangle = 0$$

$$\Rightarrow \langle X \rangle = A\cos\omega t + B\sin\omega t$$

$$\Rightarrow \langle P \rangle = m\langle \dot{X} \rangle = -m\omega A\sin\omega t + m\omega B\cos\omega t$$

Consider the initial condition :

$$\begin{cases} \langle X(0) \rangle = \sqrt{\frac{\hbar}{2m\omega}} \\ \langle P(0) \rangle = 0 \end{cases}$$

we have $A = \sqrt{\frac{\hbar}{2m\omega}}$, $B = 0$.

Therefore,

$$\begin{cases} \langle X(t)\rangle = \sqrt{\frac{\hbar}{2m\omega}}\cos\omega t \\ \langle P(t)\rangle = -\sqrt{\frac{m\hbar\omega}{2}}\sin\omega t, \end{cases}$$

which is the same as the result in Part (2).

7.4.6

Use Ehrenfest's theorem :

$$\langle \dot{a}(t)\rangle = \frac{1}{i\hbar}\langle [a, H]\rangle$$

$$= \frac{1}{i\hbar}\langle [a, (a^+a + \tfrac{1}{2})\hbar\omega]\rangle$$

$$= -i\omega\langle a\rangle$$

$$\Rightarrow \langle a(t)\rangle = \langle a(0)\rangle e^{-i\omega t}.$$

$*$ $[a, a^+] = 1$

$[a, a] = [a^+, a^+] = 0.$

$$\langle \dot{a}^+(t)\rangle = \frac{1}{i\hbar}\langle [a^+, H]\rangle$$

$$= \frac{1}{i\hbar}\langle [a^+, (a^+a + \tfrac{1}{2})\hbar\omega]\rangle$$

$$= \frac{1}{i\hbar}\cdot(-\hbar\omega)\langle a^+\rangle$$

$$= i\omega\langle a^+\rangle$$

$$\Rightarrow \langle a^+(t)\rangle = \langle a^+(0)\rangle e^{i\omega t}.$$

7.4.8

(1)

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

(2)

$$\{l_x, l_y\} = \{y p_z - z p_y,\ z p_x - x p_z\}$$

$$= \left(\frac{\partial l_x}{\partial x}\frac{\partial l_y}{\partial p_x} - \frac{\partial l_x}{\partial p_x}\frac{\partial l_y}{\partial x}\right) + \left(\frac{\partial l_x}{\partial y}\frac{\partial l_y}{\partial p_y} - \frac{\partial l_x}{\partial p_y}\frac{\partial l_y}{\partial y}\right)$$

$$+ \left(\frac{\partial l_x}{\partial z}\frac{\partial l_y}{\partial p_z} - \frac{\partial l_x}{\partial p_z}\frac{\partial l_y}{\partial z}\right)$$

$$= (0-0) + (0-0) + (p_y x - y p_x)$$
$$= x p_y - y p_x$$
$$= l_z .$$

(3) $[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y , \hat{z}\hat{p}_x - \hat{x}\hat{p}_z]$
$$= [\hat{y}\hat{p}_z , \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z , \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y , \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y , \hat{x}\hat{p}_z]$$
$$= \hat{y}[\hat{p}_z , \hat{z}]\hat{p}_x - 0 - 0 + \hat{x}[\hat{z} , \hat{p}_z]\hat{p}_y$$
$$= -i\hbar \hat{y}\hat{p}_x + i\hbar \hat{x}\hat{p}_y$$
$$= i\hbar (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)$$
$$= i\hbar \hat{L}_z .$$

7.5.2

$$a|n\rangle = \sqrt{n}\,|n-1\rangle$$
$$a = \sqrt{\frac{m\omega}{2\hbar}}\,x + i\sqrt{\frac{1}{2m\omega\hbar}}\,p$$

In X-basis : $a = \sqrt{\frac{m\omega}{2\hbar}}\,x + i\sqrt{\frac{1}{2m\omega\hbar}}\left(-i\hbar\frac{d}{dx}\right)$
$$= \frac{1}{\sqrt{2}}\left(\sqrt{\frac{m\omega}{\hbar}}\,x + \sqrt{\frac{\hbar}{m\omega}}\frac{d}{dx}\right)$$
$$= \frac{1}{\sqrt{2}}\left(y + \frac{d}{dy}\right)$$
where $y \equiv \sqrt{\frac{m\omega}{\hbar}}\,x$.

Therefore, we have
$$\frac{1}{\sqrt{2}}\left(y + \frac{d}{dy}\right)\psi_n(x) = \sqrt{n}\,\psi_{n-1}(x)$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left(y + \frac{d}{dy}\right)\alpha_n H_n(y) e^{-y^2/2} = \sqrt{n}\,\alpha_{n-1} H_{n-1}(y) e^{-y^2/2}$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left(\cancel{y\alpha_n H_n(y) e^{-y^2/2}} + \alpha_n H_n'(y) e^{-y^2/2} - \cancel{y\alpha_n H_n(y) e^{-y^2/2}}\right) =$$
$$\sqrt{n}\,\alpha_{n-1} H_{n-1}(y) e^{-y^2/2}$$
$$\Rightarrow H_n'(y) = \sqrt{2n}\,\frac{\alpha_{n-1}}{\alpha_n} H_{n-1}(y) ,$$
where
$$\frac{\alpha_{n-1}}{\alpha_n} = \left\{\frac{2^{2n}(n!)^2}{2^{2(n-1)}[(n-1)!]^2}\right\}^{\frac{1}{4}} = \sqrt{2n} .$$

Therefore, $H_n'(y) = 2n\,H_{n-1}(y)$.

7.5.3

$$a + a^{+} = \sqrt{2}\, y$$

$$(a+a^{+})|n\rangle = \sqrt{n}\,|n-1\rangle + \sqrt{n+1}\,|n+1\rangle$$

In X-basis,

$$\sqrt{2}\, y\, \psi_n(x) = \sqrt{n}\, \psi_{n-1}(x) + \sqrt{n+1}\, \psi_{n+1}(x)$$

$$\Rightarrow \sqrt{2}\, y\, \alpha_n H_n(y) e^{-y^2/2} = \sqrt{n}\, \alpha_{n-1} H_{n-1}(y) e^{-y^2/2} + \sqrt{n+1}\, \alpha_{n+1} H_{n+1}(y) e^{-y^2/2}$$

$$\because \alpha_n = \left(\frac{m\omega}{\pi\hbar\, 2^{2n}(n!)^2}\right)^{\frac{1}{4}}$$

$$\Rightarrow \sqrt{2}\, y \left(\frac{1}{2^{2n}(n!)^2}\right)^{\frac{1}{4}} H_n(y) = \sqrt{n}\left\{\frac{1}{2^{2(n-1)}[(n-1)!]^2}\right\}^{\frac{1}{4}} H_{n-1}(y) + \sqrt{n+1}\cdot \left\{\frac{1}{2^{2(n+1)}[(n+1)!]^2}\right\}^{\frac{1}{4}} H_{n+1}(y)$$

$$\Rightarrow 2y\, H_n(y) = 2n\, H_{n-1}(y) + H_{n+1}(y)$$

$$\Rightarrow H_{n+1}(y) = 2y\, H_n(y) - 2n\, H_{n-1}(y).$$